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Evaluating irrigation water demand response to climate change scenarios with the CROPWAT model in the Hemavathy command area, Karnataka

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Abstract

Climate change significantly threatens agricultural water security, especially in semi-arid regions with high cropping intensity and limited water resources. This study assesses current and future irrigation water requirements in the Hemavathy command area of Karnataka, India, using the CROPWAT 8.0 model integrated with historical meteorological data from the Indian Meteorological Department (1985–2014) and projected climate data from the MIROC5 regional climate model under RCP 4.5 and RCP 8.5 scenarios (2020–2035). Spatial interpolation of climatic variables was performed using the Thiessen polygon method to generate area-weighted averages across the command area. Gross Irrigation Requirement (GIR) was estimated based on seasonal cropping patterns and effective rainfall using the Soil Conservation Service (SCA) method. Beyond standard irrigation demand estimation, the study introduces a Water Stress Index (WSI) to quantify the balance between irrigation demand and water availability, performs sensitivity analysis of key drivers (ET_0, rainfall, crop coefficient), and evaluates adaptation scenarios including improved irrigation efficiency and crop pattern diversification. Results indicate a significant increase in irrigation demand under future climate scenarios, with projected requirements reaching up to 31.80 Thousand-Million m^3 (TMC) under RCP 4.5, as compared to 27.13 TMC in 2012–2013 under historical conditions, accompanied by higher WSI indicating elevated irrigation stress. Sensitivity analysis highlights reference evapotranspiration (ET_0) and rainfall as dominant controls on irrigation stress. Sensitivity analysis highlights ET_0 and rainfall as dominant controls on irrigation requirements, while adaptation scenarios demonstrate that integrated strategies can reduce irrigation demand by over 30% and lower WSI below the critical threshold. Analysis of reservoir inflow and groundwater yield revealed approximately 6.4 stressed years over 32 years, with groundwater alone insufficient to meet irrigation demand in most years. The study highlights the urgent need for integrated and adaptive irrigation strategies, including conjunctive water use, improved efficiency, and climate-responsive planning, to ensure agricultural sustainability under changing climatic conditions.

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Keywords Irrigation demand, Climate Change, CROPWAT 8.0, MIROC5, RCP 4.5 & 8.5, Hemavathy command

1 Introduction

Agriculture is the largest consumer of freshwater globally, accounting for nearly 70% of total freshwater withdrawals, and irrigation plays a critical role in sustaining food production and rural livelihoods, particularly in developing countries [6]. In many developing regions, particularly in South Asia, irrigation systems support highly intensive agricultural production but are increasingly vulnerable to climatic variability and water scarcity. Changes in rainfall patterns, temperature regimes, and evapotranspiration rates caused by climate change can significantly influence crop water requirements and irrigation planning [13, 21]**.** Consequently, understanding how irrigation demand may evolve under changing climatic conditions has become a major concern for sustainable water resource management [1, 15]. In India, increasing population pressure, agricultural intensification, and climate variability have significantly increased stress on available water resources [14, 16]. Semi-arid regions of southern India are especially vulnerable because agricultural productivity is highly dependent on monsoon rainfall and irrigation systems. Variations in rainfall distribution, increasing temperature, and prolonged dry spells directly influence crop water requirements and irrigation scheduling [11, 28].

Climate change is expected to substantially alter hydrological and agricultural systems through rising temperatures, shifting precipitation patterns, and increased evapotranspiration demand [13]. Even modest temperature increases can substantially increase reference evapotranspiration (ET_o), thereby raising crop water requirements and irrigation demand, particularly in semi-arid and water-stressed regions [23, 28]. In addition, variability in monsoon rainfall and increasing frequency of extreme weather events may further intensify irrigation stress and groundwater depletion. These changes pose serious challenges for agricultural water management and require improved assessment tools to evaluate future irrigation demand under different climate scenarios. Several studies conducted across India have reported that increasing temperatures and irregular monsoon rainfall may significantly increase irrigation water requirements in agricultural command areas [22, 25, 30].

To evaluate potential future climatic conditions, climate simulations derived from General Circulation Models (GCMs) and Regional Climate Models (RCMs) are widely used. RCMs provide relatively high spatial resolution and improved representation of regional climatic processes compared to coarse-resolution global models, making them suitable for regional hydrological and irrigation studies [8, 27]. Future climate projections are generally represented using Representative Concentration Pathways (RCPs), which describe different greenhouse gas concentration trajectories based on radiative forcing levels by the end of the century [13]. Among these scenarios, RCP 4.5 represents a moderate stabilization pathway, while RCP 8.5 represents a high-emission pathway associated with greater warming intensity. Tools such as CROPWAT 8.0, developed by the Food and Agriculture Organization (FAO), are widely used for estimating crop water requirements, irrigation scheduling, and effective rainfall under different climatic scenarios [5, 12, 18]. By integrating climatic variables, crop parameters, and soil characteristics, such models provide valuable insights into irrigation demand dynamics.

When combined with climate projections derived from high-resolution models such as MIROC5, these tools enable detailed evaluation of future irrigation requirements and water stress conditions.

Several researchers have applied the CROPWAT model to estimate crop water requirements and irrigation demand under changing climatic conditions. For instance, Gangwar et al. [7] used CROPWAT to estimate irrigation demand under varying climatic scenarios in northern India, while Swain et al. [28] applied the model to assess irrigation requirements in eastern India under future climate projections. Similar studies conducted in semi-arid regions of South and West Asia have reported substantial increases in irrigation demand due to rising evapotranspiration and rainfall variability [9, 15, 24]. These studies highlight the importance of integrating climate projections with crop water assessment models for sustainable irrigation planning.

Despite the increasing number of climate impact studies, relatively fewer studies have focused specifically on irrigation command areas where local-scale climatic variability, reservoir operation, cropping patterns, and groundwater interactions strongly influence irrigation demand [2, 19, 22]. Most previous studies have concentrated on basin-scale hydrological assessments, often overlooking the spatial and operational complexities of canal irrigation systems in semi-arid agricultural regions [2, 22]. Command-area scale analyses are particularly important in regions with intensive irrigation systems, where small changes in climate variables may significantly affect agricultural water use and reservoir operation strategies. Therefore, detailed command-area level assessments are essential for improving irrigation planning and climate adaptation strategies.

The Hemavathy command area in Karnataka represents an important agricultural region dependent on canal irrigation and monsoon rainfall. Crops such as paddy, maize, ragi, and pulses dominate the agricultural landscape and exhibit varying sensitivity to climatic fluctuations. Increasing climatic uncertainty combined with rising irrigation demand necessitates a comprehensive evaluation of future irrigation water requirements in the region. In Karnataka and other peninsular regions, declining reservoir inflows and increasing irrigation demand have already emerged as major concerns for sustainable agricultural water management [10, 20]. Previous studies have shown that even small increases in reference evapotranspiration (ET_o) can substantially increase crop water demand, particularly for water-intensive crops such as paddy [3, 4, 26].

In this context, the present study evaluates irrigation water demand in the Hemavathy command area using historical meteorological datasets and future climate projections derived from the MIROC5 regional climate model under RCP 4.5 and RCP 8.5 scenarios. Unlike previous large-scale basin studies, the present research provides a detailed command-area level assessment integrating climate projections, crop water requirement analysis, and irrigation demand estimation. The findings of the study are expected to support climate-resilient irrigation planning and sustainable water resource management in semi-arid agricultural regions.

The specific objectives are as follows:

- To estimate present and future irrigation water requirements using CROPWAT 8.0 under historical and projected climate scenarios.
- To analyse the impact of climate variability on evaporation, effective rainfall, and irrigation demand in the study area.

- To identify water stress years and recommend adaptation strategies for sustainable irrigation planning.

2 Study area

The study was conducted in the Hemavathy command area, located in southern Karnataka, India, served by the Hemavathy Reservoir constructed across the Hemavathy River, a tributary of the Cauvery. The command area lies between 12°30′ 00′′ N to 13°30′ 00′′ N latitude and 75° 45′ 00′′ E to 77°00′ 00′′ E longitude, covering a Culturable Command Area (CCA) of 265,074 hectares across parts of Hassan, Mandya, and Tumkur districts. The region experiences a semi-arid tropical climate, with an average annual rainfall of 800–1000 mm, predominantly during the southwest monsoon. Soils include loam, sandy clay loam, and black cotton, which support crops such as paddy, maize, ragi, and pulses. The command area follows a cropping intensity where approximately 60% of the cultivated area is under Kharif crops and 40% under Rabi crops. A section of the command area under the Hemavathy project lies within the basin, while some parts are adjacent to it, at an elevation of 1219 m. The catchment area of the Hemavathy dam extends over approximately 2810 km^2, serving a command area of 2650 km^2. The Hemavathy River extends over a distance of approximately 245 km, flowing through Hassan district, where it meets its primary tributary, the Yagachi River. It then proceeds into Mandya district, ultimately converging into the backwaters of the Krishna Raja Sagara (KRS) reservoir. Notable tributaries include the Vatehole, Yagachi, and Igur rivers. The riverbed sits at an elevation of 850.09 m, with the Full Reservoir Level (FRL) at 890.63 m and the irrigation outlet level at 867.16 m. This dam serves the singular purpose of irrigation, being a gravity dam that extends approximately 4.9 km with an altitude of about 44.5 m. Its storage capacity amounts to 37.103 TMC (Thousand-Million m^3), comprising a dead storage of 1.343 TMC and a live storage capacity of 32.731 TMC. The location of the study area is depicted in Fig. 1.

The area is irrigated through a canal network sourced from the Hemavathy dam, which has a gross storage capacity of 1050 MCM, with groundwater serving as a supplemental source during water-stressed periods. Given its reliance on rainfall and increasing water demand, the Hemavathy command area provides a critical setting to assess irrigation requirements and implement climate-resilient strategies using integrated modelling approaches.

3 Data collection and preparation

This study adopts a comprehensive approach for estimating present and future irrigation demand in the Hemavathy command area using historical data, projected climate scenarios, and crop water simulation models.

3.1 Historical meteorological datasets

Meteorological data for the historical period (1985–2014) were obtained from the Indian Meteorological Department (IMD). This dataset includes: Rainfall (0.25° resolution); Maximum and minimum temperature (1.0° resolution, rescaled to 0.25°); Relative humidity, bright sunshine hours, and wind speed [17]. Reservoir inflow data and groundwater yield data were also incorporated to assess water availability and irrigation

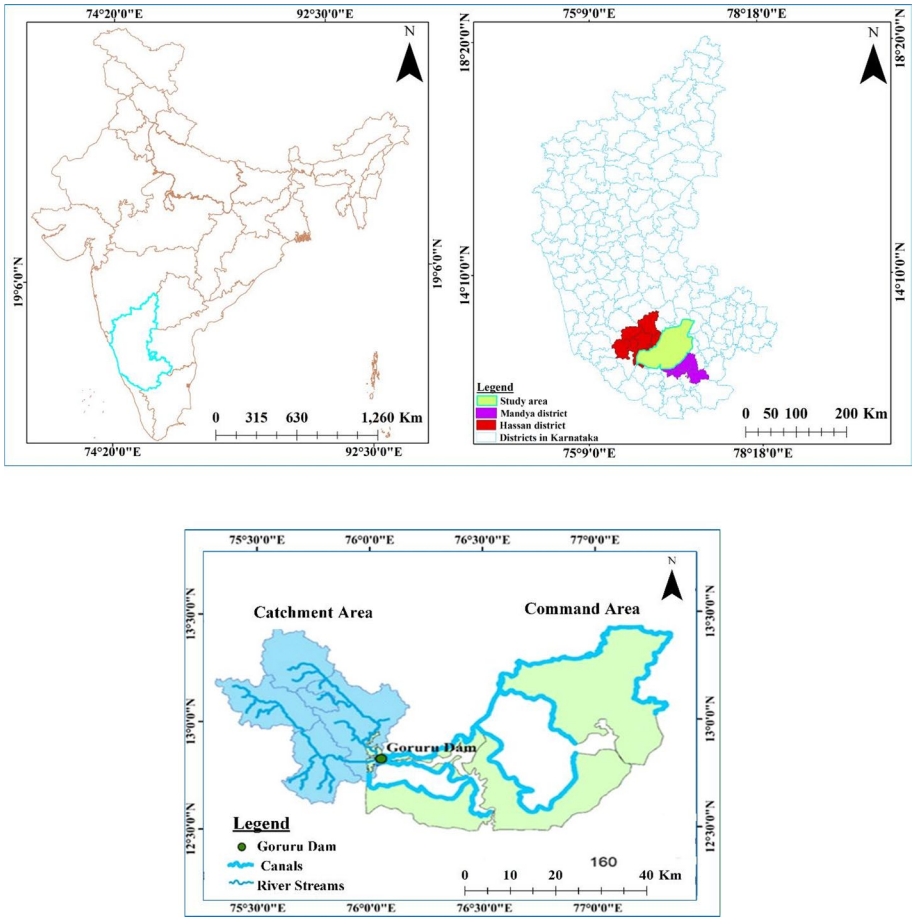


Fig. 1 Location map of the study area

stress conditions. To ensure consistency in spatial resolution, all meteorological parameters were harmonised to a 0.25° grid using interpolation and area weighting techniques.

3.2 Climate model data

To simulate future climate conditions, climate projection data from the MIROC5 (Model for Interdisciplinary Research on Climate), Regional Climate Model (RCM), developed by the University of Tokyo and associated research institutes, were used under Representative Concentration Pathways (RCP) 4.5 and 8.5 scenarios. The MIROC5 climate model forms part of the CMIP5 ensemble and is available through the Earth System Grid Federation data portal (https://esgf-node.llnl.gov/projects/cmip5/). Future climate projections were analysed for two long-term time slices: 2020–2049, representing the near-future period, and 2070–2099, representing the late-century climate scenario. In addition to these aggregated projections, annual climate data for 2020–2036 were used to simulate irrigation demand using the CROPWAT 8.0 model. The climate variables obtained from MIROC5 were originally available at a spatial resolution of 0.44° and were downscaled to 0.25° using the Inverse Distance Weighted (IDW) interpolation technique in ArcGIS 10.8 to match the resolution of the IMD datasets.

The MIROC5 model was selected for its demonstrated ability to simulate temperature and precipitation patterns over the South Asian monsoon region. Previous studies have reported that MIROC5 performs well in representing monsoon variability and

seasonal climatic trends across the Indian subcontinent [11, 15, 26], making it suitable for regional climate-impact assessments and irrigation-demand studies.

3.3 Crop and soil dataset

The cropping pattern and intensity were adopted from the Detailed Project Report (DPR) of the Hemavathy Irrigation Project. Major crops include paddy, maize, finger millet (ragi), and pulses for the Kharif and Rabi seasons. The Cultural Command Area (CCA) is 265,074 ha.

Soil types across the region are predominantly loam and sandy clay loam, with black cotton soil in paddy-dominant areas. Infiltration rates for different soil types were obtained from FAO irrigation and drainage guidelines and standard soil hydraulic property literature.

4 Methodology and data arrangement

The methodology involves data collection, spatial processing, estimation of crop water requirements using the CROPWAT 8.0 model, and scenario analysis based on Regional Climate Models (RCMs). The MIROC5 climate model from the CMIP5 ensemble was selected for this study due to its reliable performance in simulating temperature and precipitation patterns over the South Asian monsoon region. Several previous studies have reported that MIROC5 provides a comparatively better representation of monsoon variability and seasonal temperature trends in the Indian subcontinent. Furthermore, bias-corrected datasets for MIROC5 were readily available and compatible with the spatial resolution required for irrigation demand modelling in the study area [29, 31]. Although CMIP6 projections are available, the MIROC5 model remains widely used in regional climate impact assessments and provides consistent projections suitable for irrigation planning studies. The methodology flowchart is illustrated in Fig. 2, detailing the sequential processing of climatic inputs, simulation modelling and demand estimation.

4.1 Spatial interpolation and weighting

The spatial distribution of meteorological parameters was computed using the Theissen polygon method to determine the areal influence of each grid station within the study area. This method assigns weightage to each meteorological station based on the area of influence around it, ensuring that spatially distributed data (rainfall and temperature) are appropriately averaged across the basin. These data were processed and converted into monthly values for use in the CROPWAT model. The spatial weighting was performed using the following equation (Eq. 1):

$$P = \frac{\sum (A_i \times P_i)}{A} \tag{1}$$

where, 'P_i' is the parameter value at station 'i';

'A_i' is the area represented by station 'i'; and.

'A' is the total area.

The spatial extent of each station's influence was delineated using the Thiessen polygon technique in ArcGIS 10.8. This is visually represented in Fig. 3, which shows the areal coverage of each meteorological station across the study area.

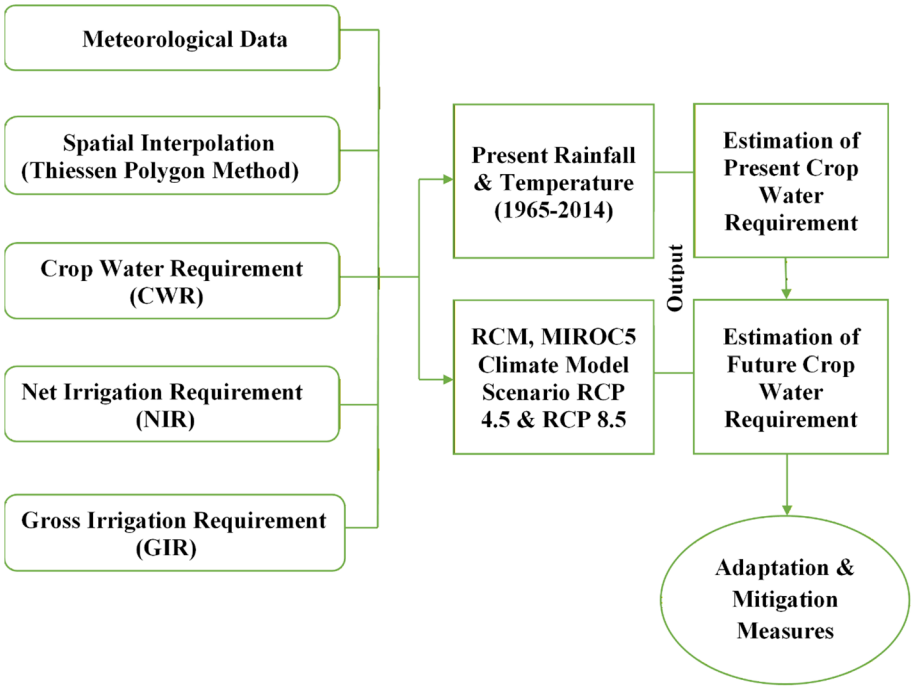


Fig. 2 Methodology framework adopted for the study

Table 1 presents each station's latitude, longitude, area (in km^2 and ha), and corresponding weightage. The calculated Thiessen polygon weights were applied to the rainfall and temperature datasets to generate spatially weighted average climatic inputs for the command area.

4.2 Effective rainfall

Effective rainfall is defined as the portion of total precipitation that is available to meet crops' water requirements after accounting for surface runoff, deep percolation, and other losses. This study estimated effective rainfall using the USDA Soil Conservation Service (SCS) method, which is built into the CROPWAT 8.0 software.

The SCS method provides a simple empirical relationship to estimate effective rainfall from (10-day) rainfall data. The formula is divided into two conditions, depending on the magnitude of rainfall over 10-day period (Eqs. 2 and 3).

- For $P_{dec} \leq 83.33$ mm (i.e., 250/3 mm):

$$P_{eff}\,(dec) = \frac{P_{dec} * (125 - 0.60 * P_{dec})}{125} \tag{2}$$

- For $P_{dec} > 83.33$ mm:

$$P_{eff}\,(dec) = \frac{(125)}{3} + 0.1 * P_{dec} \tag{3}$$

where, 'P_{eff}' = Effective rainfall during the 10-day period (mm); 'P_{dec}' = Total rainfall during the 10-day period (mm).

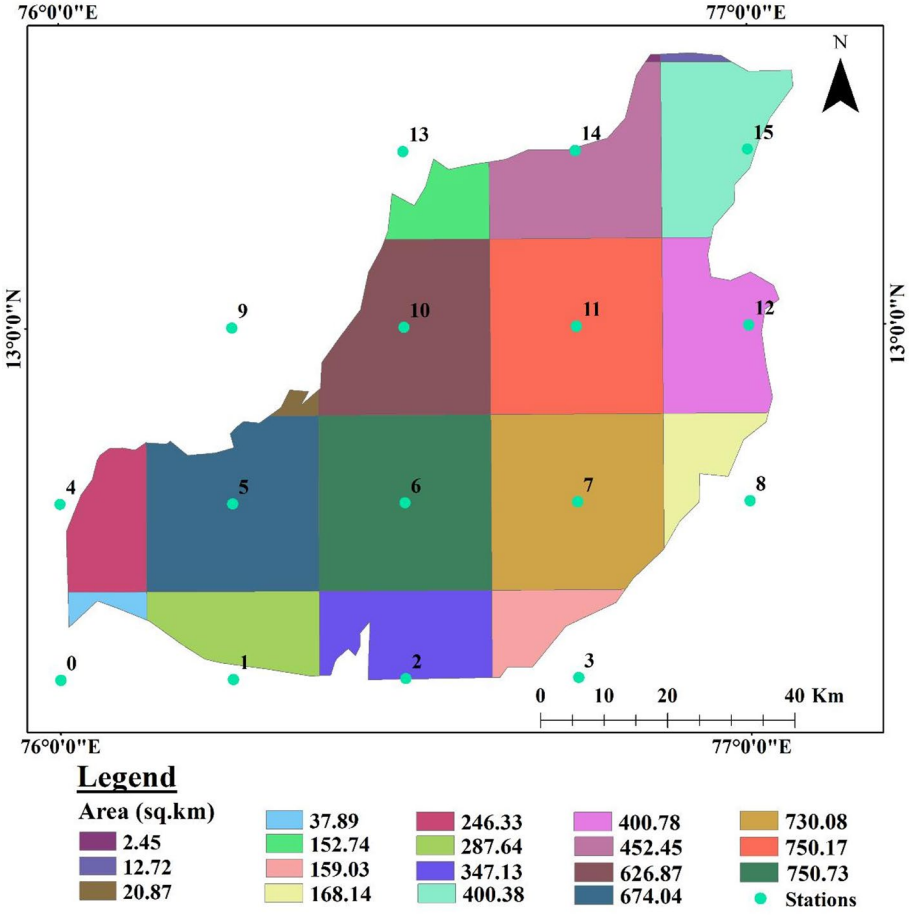


Fig. 3 Spatial distribution of thiessen polygons over the Hemavathy command area

Table 1 Thiessen polygon area and weightage for meteorological stations

S. N.	Station No.	Lat	Long	Thiessen area (km^2)	Area (Ha)	Thiessen weightage
1	0	12.50	76.00	37.89	3789	0.0061
2	1	12.50	76.25	287.63	28,763	0.0462
3	2	12.50	76.50	347.13	34,713	0.0558
4	3	12.50	76.75	159.03	15,903	0.0256
5	4	12.75	76.00	246.32	24,632	0.0396
6	5	12.75	76.25	674.05	67,405	0.1083
7	6	12.75	76.50	750.73	75,073	0.1206
8	7	12.75	76.75	730.08	73,008	0.1173
9	8	12.75	77.00	168.13	16,813	0.0270
10	9	13.00	76.25	20.87	2087	0.0034
11	10	13.00	76.50	628.87	62,887	0.1011
12	11	13.00	76.75	750.18	75,018	0.1206
13	12	13.00	77.00	400.76	40,076	0.0644
14	13	13.25	76.50	152.74	15,274	0.0245
15	14	13.25	76.75	452.42	45,242	0.0727
16	15	13.25	77.00	400.38	40,038	0.0643
17	16	13.50	76.75	2.45	245	0.0004
18	17	13.50	77.00	12.72	1272	0.0020
			Total	6222.38	622,238	

This approach is particularly suited for agricultural water management because it provides a realistic estimation of the usable portion of rainfall, especially under tropical conditions with variable rainfall intensities.

4.3 Estimation of reference evapotranspiration (ET_0)

Reference evapotranspiration (ET_0) was estimated using the Food and Agriculture Organisation (FAO) Penman-Monteith equation embedded in the CROPWAT 8.0 software. Input parameters include: Geographical latitude and longitude; Daily maximum and minimum temperature; Bright sunshine hours; Wind speed; Relative humidity.

'ET_0' was estimated for both historical and projected climate scenarios using IMD and MIROC5 datasets, respectively (Eq. 4).

$$ET_0 = \frac{0.408\Delta\ (R_n - G) + \gamma\ \left(\frac{900}{T} + 273\right) u_2 (e_s - e_a)}{\Delta\ +\ \gamma\ (1 + 0.3u_2)} \tag{4}$$

where, 'ET_o' =Reference evapotranspiration (mm day^{-1});

'R_n'= Net radiation at crop surface ($MJm^{-2}day^{-1}$);

'G' = soil heat flux density ($MJm^{-2}day$);

'T'= Mean daily temperature at 2m height(ms^{-1});

'e_s' = Saturation vapour pressure (kPa);

'e_a' = actual vapour pressure(kPa);

'$(e_s - e_a)$' = saturation vapour pressure deficit (kPa);

'Δ'= slope vapour pressure (kPa°C); and

γ' = Psychrometric constant (KPa°C).

4.4 Cropping pattern and soil parameters

The command area supports both Kharif and Rabi cropping seasons, with the Culturable Command Area (CCA) totalling 265,074 ha. Irrigation intensity is assumed to be 60% during Kharif and 40% during Rabi.

The major crops grown in the region include paddy, maize, finger millet (ragi) and pulses. Crop-specific input parameters, such as effective root zone depth and crop coefficient (K_c), were incorporated into the CROPWAT 8.0 model to estimate crop evapotranspiration (ET_c). Paddy is primarily cultivated in areas with black cotton soil, whereas other crops are grown on loam and sandy clay loam soils. The details of the cropping pattern are shown in Table 2.

Table 2 Cropping pattern and intensity of irrigation in the command area

S.N.	Crops	Intensity of irrigation (%)	Area (ha)
Kharif (60%)			
1	Paddy	2	5261
2	Ragi	20	53,895
3	Maize	20	52,355
4	Pulses	18	47,736
Rabi (40%)			
1	Ragi	14	37,040
2	Maize	14	35,980
3	Pulses	12	32,807
	Total	100	265,074

Department of Agriculture, Government of Karnataka, 2010–2011

Table 3 Crop Coefficient (K_c) of Major Crops

CROPS	Kc			
	Land preparation	Initial	Middle	Late
Paddy (Dry)	0.30	0.50	1.05	0.70
Paddy (Wet)	1.05	1.10	1.20	1.05
Maize	-	0.30	1.20	0.35
Finger Millet(Ragi)	-	0.30	1.00	0.30
Pulses	-	0.40	1.15	0.35

Table 4 Infiltration rates of soil

Soil type	Value of infiltration rate (cm/hour)
Clay Loam	0.60–0.80
Sandy Loam	1.20–1.80

$$ET_c = ET_0 * K_c \tag{5}$$

Crop evapotranspiration (ET_c) is calculated as the product of reference evapotranspiration (ET_0) and crop coefficient (K_c), which varies depending on the crop's growth stage. Two sets of 'K_c' values for paddy are considered: dry and wet conditions, reflecting field preparation and irrigation practices (Table 3). The general equation used is (Eq. 5):

The soil in command is classified into loam, sandy loam, and black cotton soil. These medium-textured soils are well-suited for crop production. Infiltration rate, a key input for irrigation scheduling, was defined for each soil type based on standard literature values, ranging from 0.6 to 1.8 cm/hr depending on texture (Table 4).

4.5 Crop water requirement (CWR)

The Crop Water Requirement (CWR) for each crop is then derived by summing the daily crop evapotranspiration across the growing season, while also accounting for land preparation needs and percolation losses (Eq. 6):

$$CWR = ET_c + W_{lp} + D_p \tag{6}$$

where, 'ET_c' = Crop Evapotranspiration.

'W_{lp}' = Water for land preparation.

'D_p' = Deep percolation losses.

The effective rainfall, calculated using the USDA Soil Conservation Service (SCS) method, which is built into the CROPWAT model, is subtracted from the CWR to determine the Net Irrigation Requirement (NIR), which is the portion of the water requirement that must be met through artificial irrigation.

4.6 Estimation of net and gross irrigation requirements

The estimation of irrigation requirements is a critical component of water resources management, especially in command areas facing climatic variability. In this study, both Net Irrigation Requirement (NIR) and Gross Irrigation Requirement (GIR) were estimated for existing and projected climate scenarios using the CROPWAT 8.0 model.

4.6.1 Soil-water relationship

Irrigation planning is strongly influenced by the soil type, crop characteristics and root zone moisture dynamics. Over-irrigation may lead to waterlogging and reduced crop

yields, while under-irrigation can result in moisture stress. Hence, accurate water balance modelling in the root zone is essential.

The NIR represents the volume of water needed to compensate for the deficit between crop evapotranspiration (ET_c) and effective rainfall (P_{eff}), excluding negligible contributions from groundwater. It is calculated using the following equation (Eq. 7):

$$NIR = TWR + Percolation\ losses - P_{eff} - GW \tag{7}$$

where, '*NIR*' = Net Irrigation Requirement (mm);

'*TWR*' = Total Water Requirement (mm);

'P_{eff}' = Effective Rainfall (mm);

'*GW*' = Groundwater contribution (considered negligible in this study).

The daily soil moisture balance for each crop is simulated using the CROPWAT 8.0 model based on crop root zone depth. Irrigation scheduling is based on the principle that once more than 50% of the available soil moisture is depleted, yield begins to decline significantly. This threshold is supported by previous field studies on cotton, maize and sorghum under Indian agro-climatic conditions[32].

4.6.2 Gross irrigation requirement (GIR)

The Gross Irrigation Requirement (GIR) accounts for various system inefficiencies, including field application and conveyance losses. It is computed in two steps (Eqn: 8 & 9):

1. Field Irrigation Requirement (FIR):

$$FIR = \frac{NIR}{Field\ Efficiency} \tag{8}$$

- For rice, field efficiency = 90%.
- For non-rice crops, field efficiency = 70%.

2. Gross Irrigation Requirement (GIR):

$$GI = \frac{FIR}{System\ Efficiency} \tag{9}$$

- System efficiency (accounting for conveyance, field and operational losses) = 75%.

Thus, the GIR reflects the total water required at the irrigation headwork to ensure adequate supply at the field level.

Projected irrigation demands were estimated for future time slices using the MIROC5 climate model outputs under RCP 4.5 and RCP 8.5. The results were compared with historical demands to assess the impact of climate change on irrigation water requirements.

4.7 Estimation of crop water requirement for present and future scenarios

To evaluate the impact of climatic variability on irrigation demand, crop water requirements were estimated separately for the present scenario (1985–2014) and future scenarios (2020–2035) using historical IMD data and MIROC5 climate projections under RCP 4.5 and RCP 8.5.

The CROPWAT 8.0 model computed seasonal crop water requirements by incorporating climate parameters such as rainfall, temperature, bright sunshine hours, wind

speed and relative humidity. The Penman-Monteith equation was applied to estimate reference evapotranspiration (ET_0), and crop-specific coefficients (K_c) were used to compute evapotranspiration (ET_c). Effective rainfall was subtracted to derive each scenario's net irrigation requirement (NIR).

This approach allowed the study to compare current and projected irrigation demands and informed the identification of adaptation and mitigation strategies. These include optimising cropping patterns, enhancing water-use efficiency, and promoting conjunctive use of surface and groundwater resources to address sustainability under changing climate conditions.

5 Results and discussions

The results are derived from IMD historical datasets (1985–2014) and climate projections from the MIROC5 model under RCP 4.5 and RCP 8.5 scenarios. The data reveal significant changes in both the timing and intensity of rainfall and a consistent warming trend in both maximum and minimum temperatures across all future years. These changes directly impact evapotranspiration, effective rainfall, and consequently, the irrigation water requirements.

5.1 Historical climate trends (1985–2014)

Historical data (1985–2014) shows October as the peak rainfall month, followed by August and September. However, under MIROC5 projections for 2020–2036, a temporal shift is observed with August emerging as the dominant peak month. Figure 4 illustrates this seasonal shift, along with inter-annual variability in total monthly rainfall. This shift has important implications for crop water scheduling, as it compresses the effective rainfall window and may increase the demand for supplementary irrigation, especially in the late Kharif and early Rabi seasons.

The historical monthly mean maximum temperature (T_{max}) ranged from 27.70 °C to 28.50 °C in 2010–2015. However, MIROC5 projection shows a consistent warming trend, with summer months (April-May) reaching up to 34 °C and a clear rise in maximum temperature across all months (Fig. 5). The Rabi season months, which were previously cooler, now exhibit significantly higher maximum temperatures. These shifts

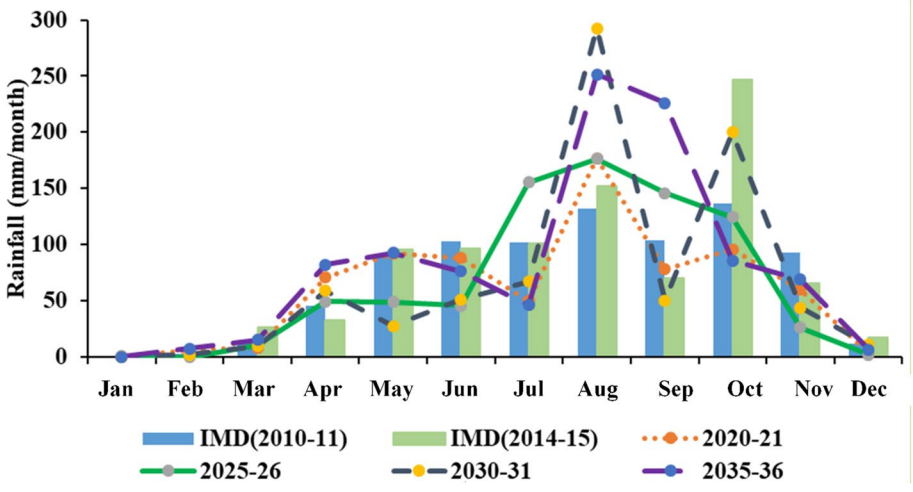


Fig. 4 Monthly Total Rainfall: Historical IMD vs. MIROC5 RCP Projected Scenario

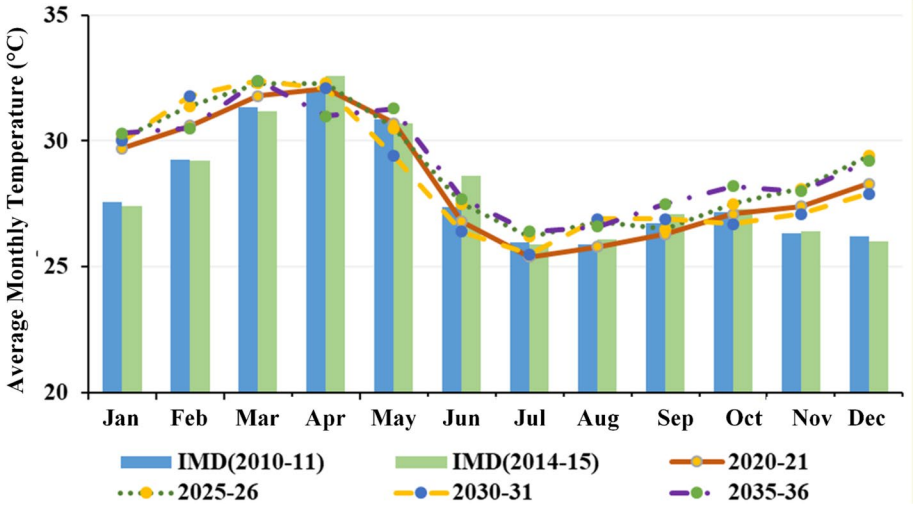


Fig. 5 Monthly mean maximum temperature: historical IMD vs. MIROC5 projected scenario

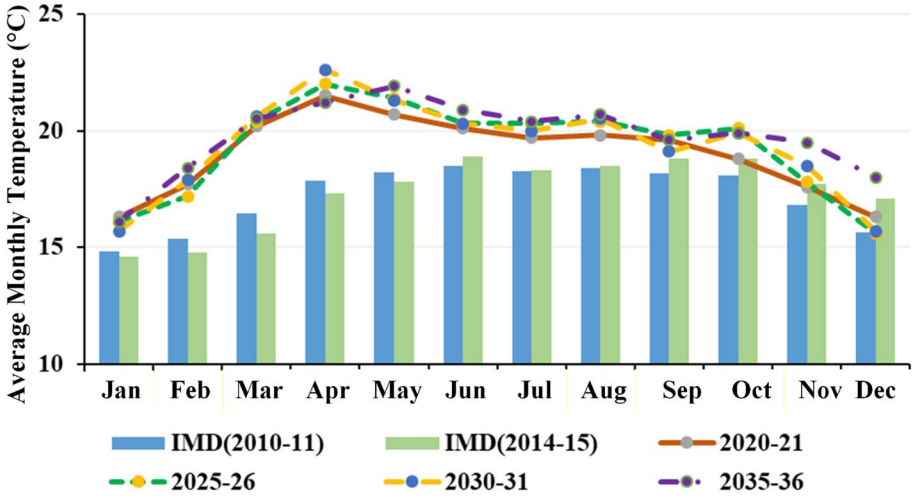


Fig. 6 Monthly mean minimum temperature: historical IMD vs. MIROC5 projected scenario

suggest higher crop water demand due to increased evapotranspiration, particularly during late Kharif and Rabi seasons.

Minimum temperatures also show a consistent warming pattern. While IMD data for 2010–2015 indicated average monthly T_{min} values of 17.80 °C to 18.00 °C, MIROC5 projections show an upward shift of 1.50 °C to 2.00 °C, especially in the cooler months (October to February). Figure 6 illustrates that projected T_{min} in post-monsoon and winter months is consistently higher than historical values, which could reduce nighttime cooling and impact soil moisture retention. This has implications for crop development stages such as flowering and grain filling, especially for Rabi crops.

Figure 7 compares the monthly rainfall distribution from the IMD time series (1985–2014) with projected rainfall under MIROC5 RCP 4.5 for the periods 2020–2049 and 2070–2099. The results indicate a temporal shift in rainfall intensity and seasonality. While October was the peak rainfall month in historical records, August emerges as the peak rainfall month in future projections. This shift has implications for irrigation scheduling, crop calendar planning, and reservoir operation strategies.

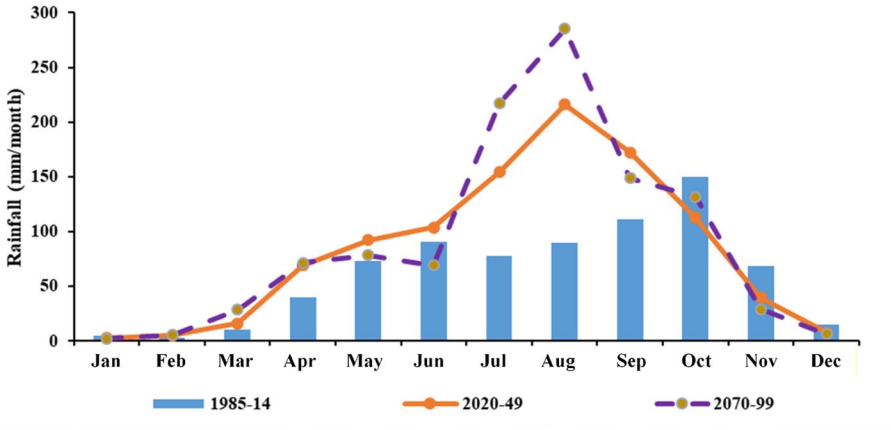


Fig. 7 Comparison of Monthly Rainfall (1985–2014)-IMD vs. MIROC5 RCP 4.5 Projections (2020–2049 & 2070–2099)

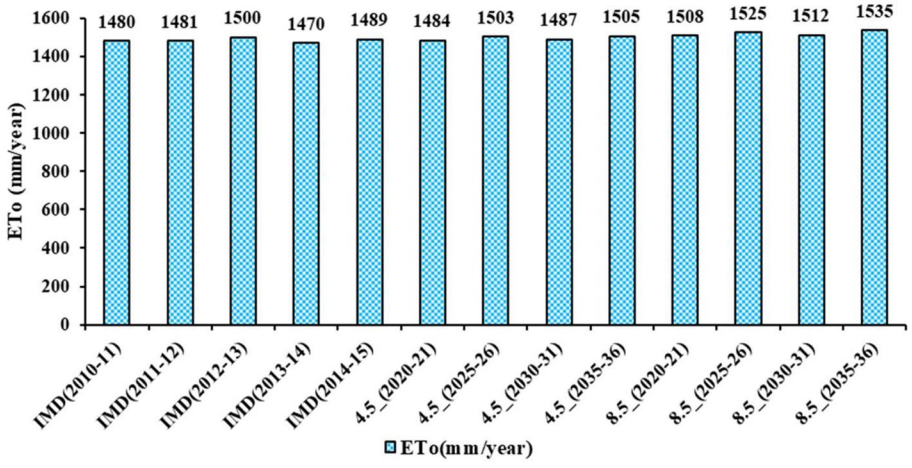


Fig. 8 Annual Reference Evapotranspiration (ET_0) Using IMD and MIROC5 Data Under RCP 4.5 & 8.5 Scenarios

5.2 Present scenario: irrigation demand estimation

Figure 8 illustrates the annual reference evapotranspiration (ET_0) in millimetres per year for both historical and projected future climate scenarios (MIROC5 under RCP 4.5 & 8.5). The data spans from 2010 to 2011 to 2035–2036 and includes projections for three future periods under each scenario.

In the historical period (2010–2015), ET_0 values ranged from 1470 mm/year to 1500 mm/year, with the highest value observed in 2012–2013, correlating with a drought year and high irrigation demand. In the projected periods under RCP 4.5, ET_0 gradually increases from 1484 mm/year (2020–2021) to 1505 mm/year (2035–2036). A more pronounced rise is seen under RCP 8.5, where ET_0 climbs from 1508 mm/year (2020–2021 to a peak of 1535 mm/year (2035–2036).

These findings confirm the anticipated rise in evaporative demand due to global warming and climate change. Higher ET_0 implies increased crop water requirements, especially during the dry season, and reduced effective rainfall contribution. This reinforces the projection that future irrigation water demand will intensify, particularly under high-emission scenarios like RCP 8.5.

In the present scenario, Fig. 9 presents the Net and Gross Irrigation Requirements (NIR & GIR) for major crops, including paddy, ragi, maize and pulses, during Kharif and

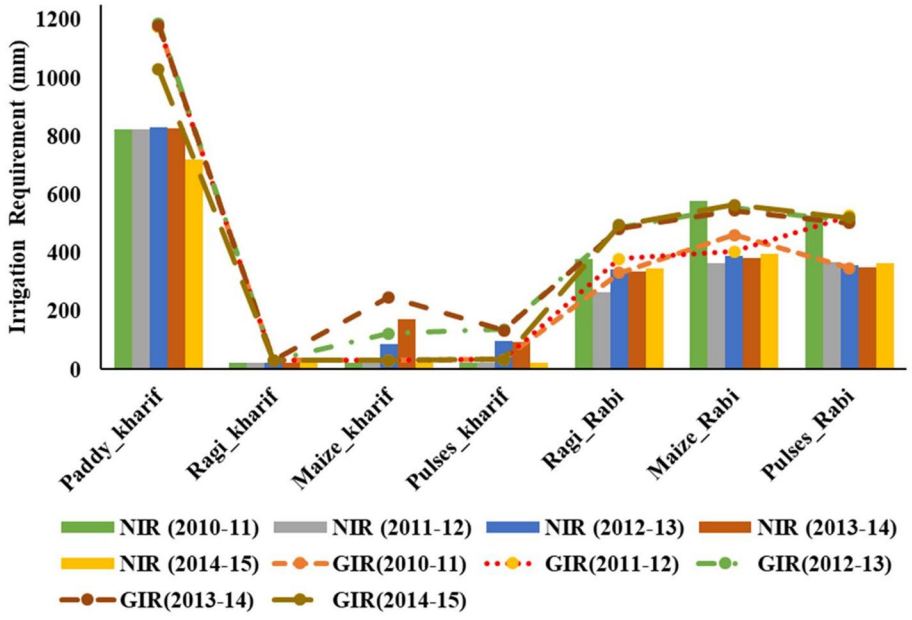


Fig. 9 Net and Gross Irrigation Requirement (NIR & GIR) for major crops in Kharif and Rabi Seasons using IMD historical data

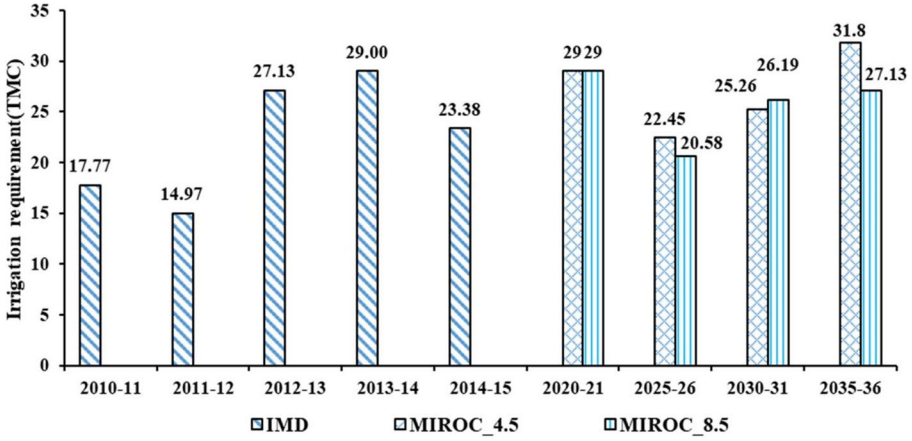


Fig. 10 Irrigation Demand for IMD Series Data (2010–2014) and MIROC5 Projections Under RCP 4.5 & 8.5 Scenarios

Rabi seasons for 2010 to 2015. The highest irrigation requirement is observed for paddy in Kharif, with GIR exceeding 1000 mm. Among Rabi crops, maize and ragi show consistent water demand across all years. These values establish the baseline irrigation burden based on IMD data.

Future projections under MIROC5 climate scenarios indicate a noticeable increase in irrigation water demand. Figure 10 illustrates the Gross Irrigation Requirement (GIR) expressed in terms of irrigation demand (TMC) for both historical IMD data (2010–2014) and projected MIROC5 scenarios. A clear increasing trend is observed, with projected irrigation demand reaching 31.80 TMC in 2035–2036 under the RCP 4.5 scenario, exceeding the historical maximum of 27.13 TMC recorded in 2012–2013. In contrast, 2010–2011 exhibited the lowest irrigation demand of 17.77 TMC, primarily due to favourable rainfall conditions and slightly lower evapotranspiration values. Further

analysis of the historical IMD dataset shows that the highest irrigation demand (29 TMC) occurred in 2013–2014 despite relatively lower ET_0 values, highlighting the strong influence of rainfall variability and cropping conditions. The lowest ET_0 (1480 mm/year) was recorded in 2013–2014, while the highest ET_0 (1500 mm/year) occurred in 2012–2013. These findings indicate that irrigation demand in the command area is not governed solely by evapotranspiration but is influenced by the combined interaction of ET_0, rainfall availability, and crop water requirements.

5.3 Projected scenario: irrigation demand under climate change

Table 5 summarises the projected climatic variables and irrigation demand under MIROC5 RCP 4.5 and RCP 8.5 scenarios for selected years. The results indicate that ET_0 generally increases under both climate scenarios due to rising temperature conditions. Under the RCP 4.5 scenario, irrigation demand increases from 29.00 TMC in 2020–2021 to 31.80 TMC by 2035–2036, mainly due to higher evapotranspiration and reduced rainfall.

In contrast, under the RCP 8.5 scenario, although ET_0 reaches its highest value (1535 mm/year), the irrigation demand decreases to 27.13 TMC in 2035–2036 because of higher rainfall availability. This is attributed to the relatively high rainfall intensity (959.70 mm) projected for that year, which reduced net irrigation needs. Conversely, in 2020–2021, although ET_0 was slightly lower (1508 mm/year), the irrigation demand rose to 29 TMC due to reduced rainfall (729.70 mm). These findings indicate that irrigation demand is influenced by the combined interaction of evapotranspiration and rainfall variability rather than evapotranspiration alone.

The processed climate data, after spatial rescaling and weighting, were applied in the CROPWAT model to estimate crop-wise Net Irrigation Requirement (NIR) and Gross Irrigation Requirement (GIR) under the MIROC5 RCP 4.5 climate scenario for selected future years (2020-21, 2025-26, 2030-31 and 2035-36). The results are presented in Fig. 11. The projected variation in irrigation requirements for major crops grown during the Kharif and Rabi seasons. Among all crops, paddy during the Kharif season exhibits the highest irrigation demand, with both NIR and GIR values significantly higher than those of other crops due to its water-intensive cultivation practices. In contrast, maize, pulses and ragi during the Kharif season show comparatively lower irrigation requirements, reflecting their relatively lower water demand. For the Rabi season, irrigation requirements increase moderately for maize, pulses and ragi, with pulses showing comparatively higher irrigation demand among other crops. The trend across the projected years indicates slight variations in both NIR and GIR values, which can be attributed to changes in projected rainfall and evapotranspiration under future climate conditions. Overall, the results highlight that crop type and seasonal cultivation patterns play a crucial role in determining irrigation demand in the study area.

Table 5 Projected irrigation demand under MIROC5 climate scenario

Scenario	Year	ET_0 (mm/year)	Rainfall (mm)	GIR (mm)	Irrigation Demand (TMC)	T_{max} (°C)	T_{min} (°C)
RCP 4.5	2020–2021	1484	908.70	3006.40	29.00	28.43	15.97
RCP 4.5	2035–2036	1505	720.90	3319.30	31.80	29.03	15.63
RCP 8.5	2020–2021	1508	729.70	3218.00	29.00	28.50	19.03
RCP 8.5	2035–2036	1535	959.70	2861.10	27.13	29.09	19.76

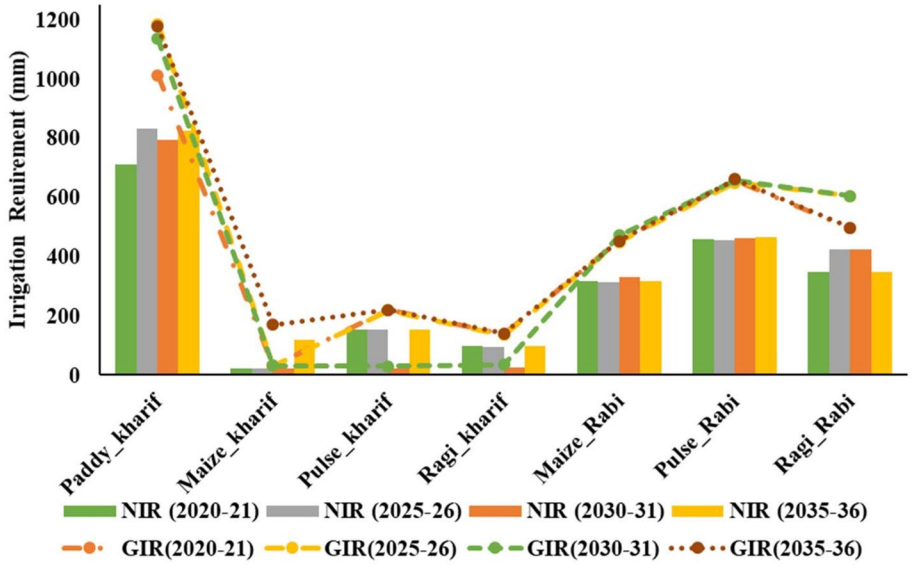


Fig. 11 Projected Net and Gross Irrigation Requirements for major crops under MIROC5 RCP 4.5 Scenario (2020–2036)

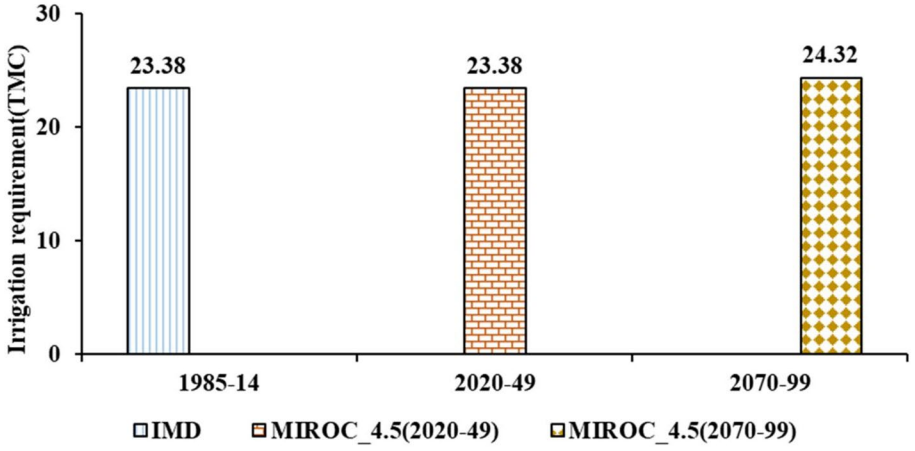


Fig. 12 Average Irrigation Demand for IMD Historical Series (1985–2014) and MIROC5 RCP 4.5 Projections (2020–2049 & 2070–2099)

5.4 Long-term comparison of demand trends

To examine broader trends, Fig. 12 compares decadal-average irrigation demand under historical and projected scenarios. While the average GIR remain steady at 23.38 TMC (baseline) for both 1985–2014 and 2020–2049, it increases slightly to 24.32 TMC by 2070–2099. Although modest, this increase underscores a gradual upward shift in water requirement due to climate change.

5.5 Water availability and stress years

A critical component of irrigation planning is assessing whether the annual inflow to the reservoir can meet the seasonal irrigation demand, after considering other competing uses such as drinking water, tank filling, downstream ecological flows, and evaporation losses.

The Hemavathy reservoir inflow data were analysed over 31 years (1985–2016) to identify years where water availability was inadequate to meet the gross irrigation requirement (GIR). The comparison of reservoir inflow versus irrigation demand, illustrated in Fig. 13, clearly identifies 6.4 water-stressed years, where the net available water in the reservoir after non-irrigation allocations was insufficient for crop irrigation. The annual reservoir inflow across this period varied significantly, ranging from as low as 31.3 TMC in 2004–2005 to as high as 139.10 TMC in 2006–2007. The orange line representing average irrigation demand (~ 27 to 29 TMC) highlights several stress years, particularly 1986–1987, 2003–2004 and 2012–2013, when available water fell short of irrigation needs. The stress years typically occurred during periods with: below-average monsoonal rainfall; high crop water demand (due to elevated ET_0); greater water allocations to other uses.

The visual trend highlights a few key years where the inflow fell significantly short of irrigation demand. Such years represent critical stress points for the command area and necessitate the implementation of contingency irrigation plans, such as reduced irrigation intensity or crop area; temporary crop shifts to less water-intensive varieties; use of stored groundwater or conjunctive irrigation strategies.

5.6 Groundwater yield vs. irrigation demand

Groundwater plays a significant role in supplementing surface water sources for irrigation, especially during dry periods or when reservoir inflows are insufficient. To evaluate its reliability, a comparison was made between estimated groundwater storage (△GWS) and irrigation demand for selected years (Fig. 14). While the groundwater storage exceeded irrigation demand in 2010–2014, it was significantly lower than irrigation demand in 2011–2013. These findings reveal that the demand consistently surpassed the available groundwater. This shortfall indicates that groundwater alone is not a dependable source for meeting irrigation needs throughout the command area.

The mismatch between groundwater yield and irrigation demand highlights several key points:

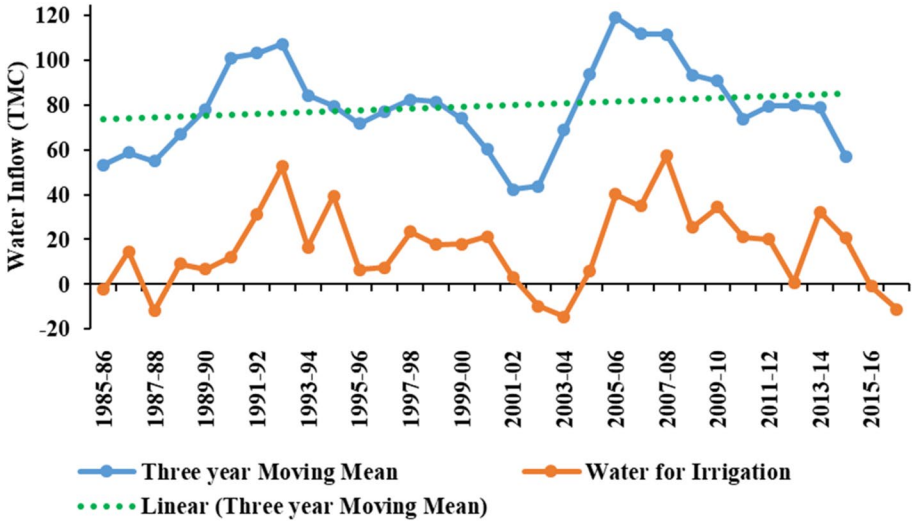


Fig. 13 Annual Reservoir Inflow and Identification of Water Stress Years in the Study Area

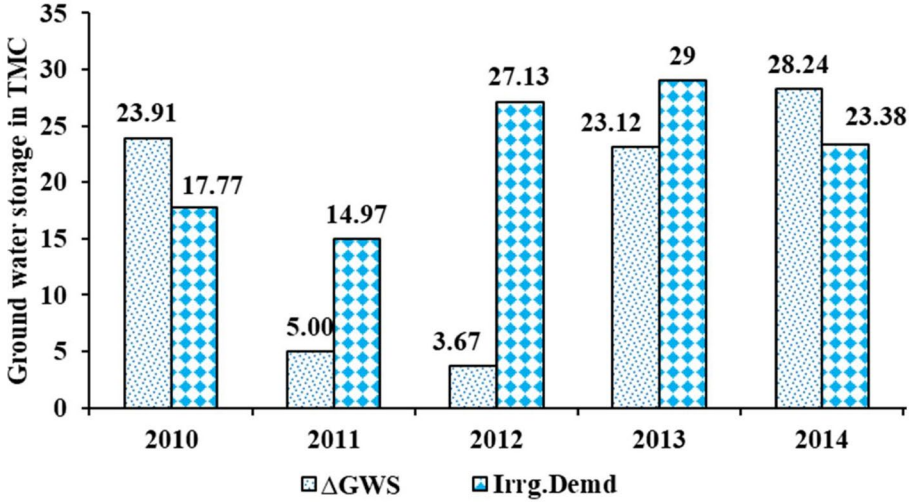


Fig. 14 Comparison of Groundwater Yield and Irrigation Demand in the Study Area

- Inter-annual variability in groundwater recharge affects its availability for agricultural use.
- In deficit years, exclusive reliance on groundwater could lead to over-extraction, threatening long-term aquifer sustainability.
- The temporal mismatch between groundwater recharge (post-monsoon) and peak irrigation demand (often pre-monsoon or post-monsoon) further complicates resource planning.

6 Conclusion

This study presents a comprehensive assessment of irrigation water demand under changing climatic conditions for the Hemavathy command area using historical and projected climate data. The results indicate an increasing trend in irrigation requirements under future climate scenarios, primarily driven by rising evapotranspiration (ET_o) and changing rainfall patterns. The findings further reveal that irrigation demand in the command area is highly sensitive to climatic variability, particularly fluctuations in temperature and rainfall distribution, which are expected to intensify under future climate conditions.

During the baseline period, the highest irrigation demand occurred in 2012–2013 (27.13 TMC), primarily due to low rainfall and elevated ET_0. In comparison, future projections indicate an increase in water demand, with the year 2035–2036 under 4.5, suggesting a requirement of 31.8 TMC. Interestingly, despite higher ET_0 in some future years, increased rainfall intensity partially mitigated irrigation stress, highlighting the complex interaction between climatic variables. Analysis of reservoir inflow reveals declining trends, with several historical years already showing shortfalls. Although groundwater contributions are significant in some years, they are insufficient to fully bridge the supply-demand gap, particularly during dry spells.

Analysis of reservoir inflows revealed approximately 6.4 years of water stress across 32 years, wherein the net available water fell short of irrigation demands after fulfilling drinking, ecological and evaporation losses. Groundwater availability was also found to be insufficient in most years, except 2010 and 2014, thereby underscoring the unreliability of a single water source.

These findings collectively reinforce the urgent need for adaptive water management strategies. This includes:

- Promotion of conjunctive use of surface and groundwater.
- Adoption of efficient irrigation technologies such as drip and sprinkler systems.
- Revised cropping patterns that are less water-intensive.
- Strengthened reservoir operation and storage planning, and.
- Development of early warning and decision-support tools based on climate projections.

In conclusion, the study quantifies the future irrigation demand under evolving climate scenarios and provides a replicable framework for integrated water resource planning. The insights generated can support policy interventions and planning decisions to enhance irrigation systems' sustainability and resilience in semi-arid and climate-sensitive regions.

Acknowledgements
The authors wish to express their sincere gratitude to the Department of Water Resources Development and Management (WRD&M), Indian Institute of Technology (IIT) Roorkee, for providing institutional support and access to essential tools and resources necessary for the successful completion of this research. We also extend our thanks to the Indian Meteorological Department (IMD) for supplying the historical climatic dataset and to the developers of the MIROC5 climate model for future projections. The guidance and insights provided by our collaborators and reviewers were instrumental in refining the scope and depth of this study. Any opinions, findings, and conclusions expressed in this paper are those of the authors and do not necessarily reflect the views of the affiliated institutions.

Author contributions
TDV led the study's conceptualization and overall research design. RK and DK were responsible for field data collection. Data acquisition, processing and the preparation of the initial manuscript draft were carried out by RK. TDV and DK contributed to data analysis and the visual presentation of findings. The manuscript was reviewed and redefined by DK and PM. Study oversight and coordination were jointly managed by RK and PM.

Funding
The authors received no specific funding or research grant for this work from any agency in the public, commercial, or not-for-profit sectors that could have influenced its outcome.

Data availability
The historical meteorological data used in this study were obtained from the Indian Meteorological Department (IMD), and the future climate projections were sourced from the Regional Climate Model, developed by the University of Tokyo. Due to licensing and access restrictions, these datasets are not publicly available; however, they can be obtained from the respective agencies upon request. Processed and derived data supporting the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable as the study does not involve human participants or animals.

Consent for publication
Not Applicable.

Competing interests
The authors declare no competing interests.

Received: 18 December 2025 / Accepted: 19 June 2026
Published online: 24 June 2026

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